// MATHEMATICS LAB · GNU OCTAVE

Mapping the Invisible

Wi-Fi Signal Mapping and Router Placement

Vector calculus applied to wireless signal strength

$$S(x,y) = 100 \cdot e^{-((x-2)^2+(y-2)^2)/10} + 80 \cdot e^{-((x-8)^2+(y-7)^2)/10}$$

100.18	0.17	61 m²	19.27
peak at (2,2)	dead zone (0,10)	usable, S > 30	path average

LAB EXPERIMENTS USED

- Exp 2 · Surface & contour
- Exp 3 · Maxima / minima
- Exp 5 · Double integral
- Exp 7 · Gradient arrows
- Exp 8 · Line integral

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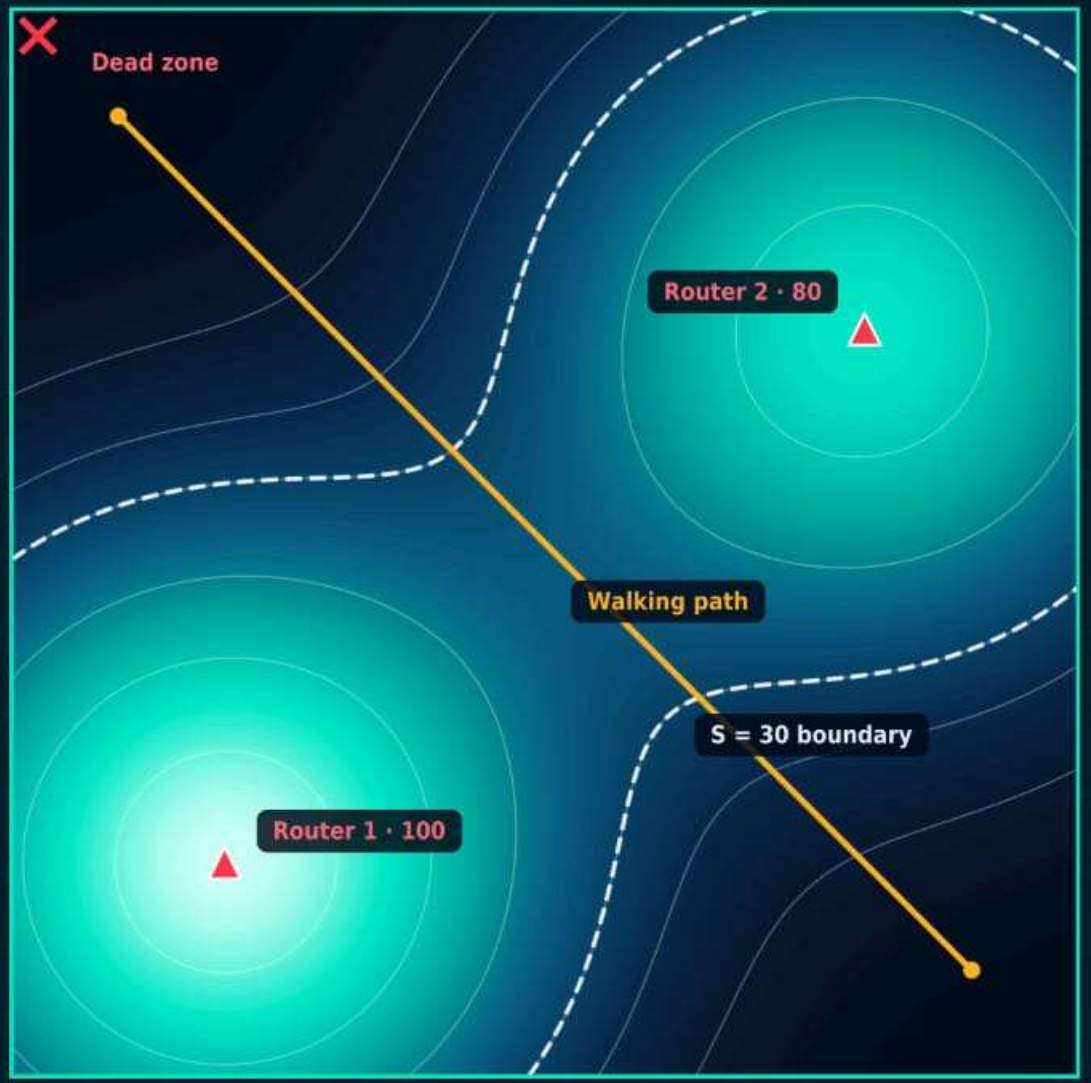
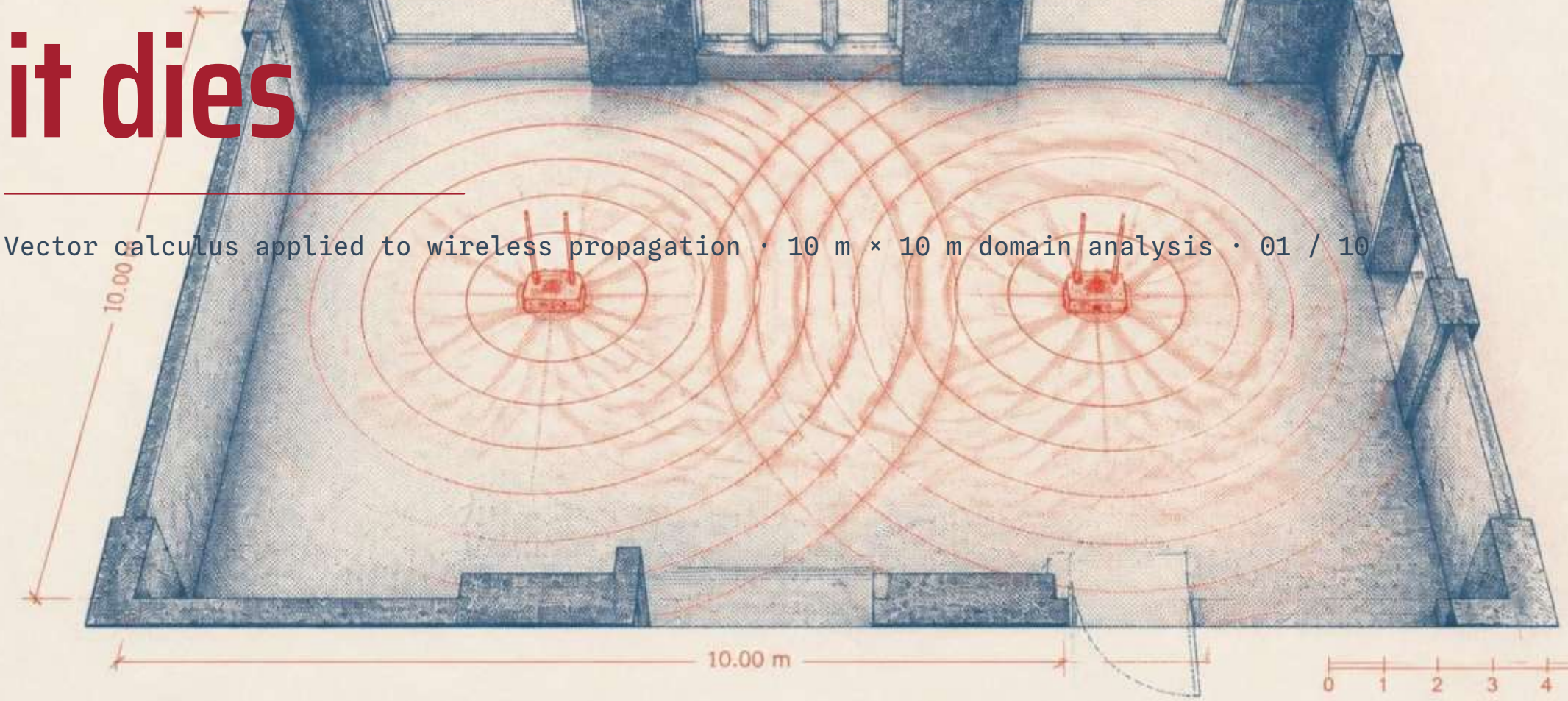


Fig. · Signal landscape S(x, y), 10 m × 10 m room

INSIDE: Signal model · Octave code · Output · Conclusion

Mapping the invisible: where Wi-Fi wins and where it dies

Vector calculus applied to wireless propagation · 10 m × 10 m domain analysis · 01 / 10



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One room, two routers, one dead corner

- 10 m × 10 m room; Router 1 at (2,2), 100 peak strength units; Router 2 at (8,7), 80 peak strength units.
- Signal fades with distance from each router.
- Dead corner at (0,10); student diagonal walk from hostel (1,9) to class (9,1) crosses weak signal and deep signal loss.
- Router placement is usually guesswork.
- Goal: place routers using calculus.

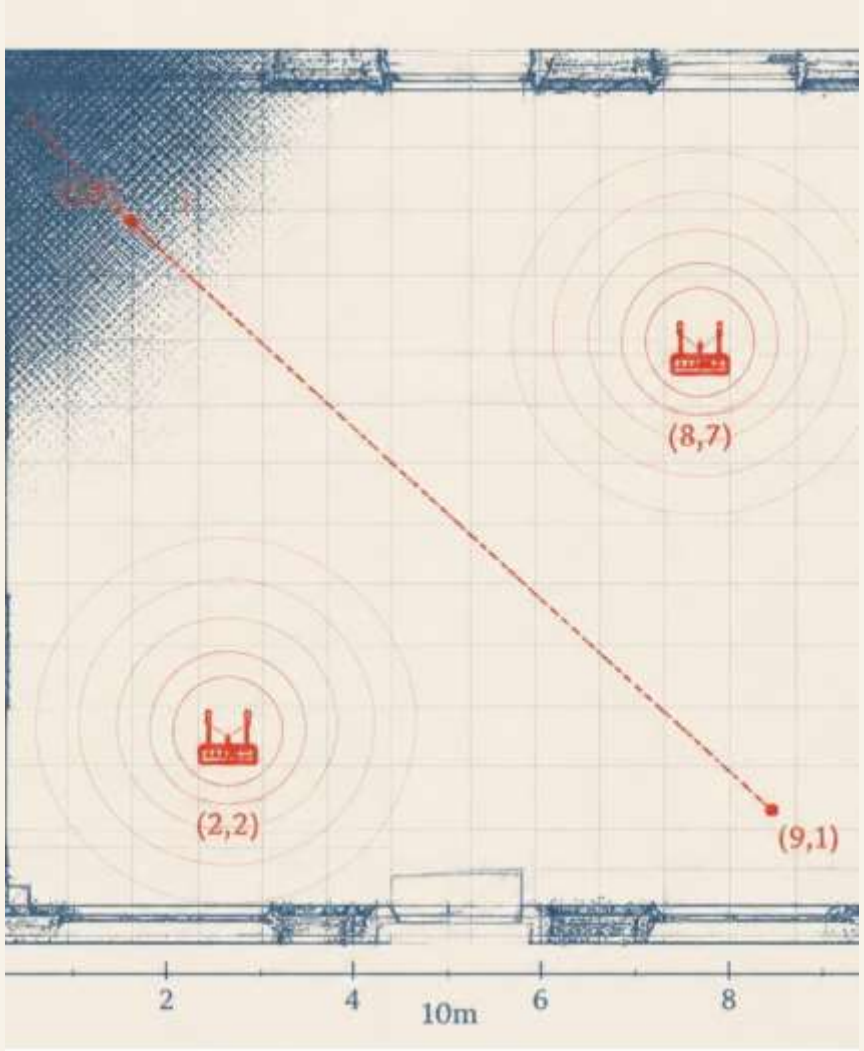


Fig. 1 · Room coordinates (0,0) to (10,10) with transmitter nodes

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One equation turns a room into a signal landscape

$$S(x,y) = 100\,e^{-\frac{(x-2)^2+(y-2)^2}{10}} + 80\,e^{-\frac{(x-8)^2+(y-7)^2}{10}}$$

Gaussian kernel

Signal decays with squared Euclidean distance scaled by decay variance 10.

Linear superposition

Dual sources sum directly at every interior spatial coordinate point.

Discrete meshgrid

Sampled over 101 × 101 nodes at uniform 0.1 m grid resolution.

```
[x, y] = meshgrid(0:0.1:10, 0:0.1:10);
S = 100*exp(-((x-2).^2 + (y-2).^2)/10) + 80*exp(-((x-8).^2 + (y-7).^2)/10);
```

Analytical model formulation · Continuous scalar field definition S: $\mathbb{R}^2 \to \mathbb{R}$ · 03 / 10

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Five calculus experiments, one pipeline

01

Surface & contour

Map the 3D topology and 2D equipotential lines across room.

02

Extrema bounds

Isolate global maximum at router and absolute floor in corner.

03

Double integral

Compute total signal volume across 100 m^2 room floor.

04

Gradient field

Determine steepest ascent vectors ∇S directing router moves.

05

Line integral

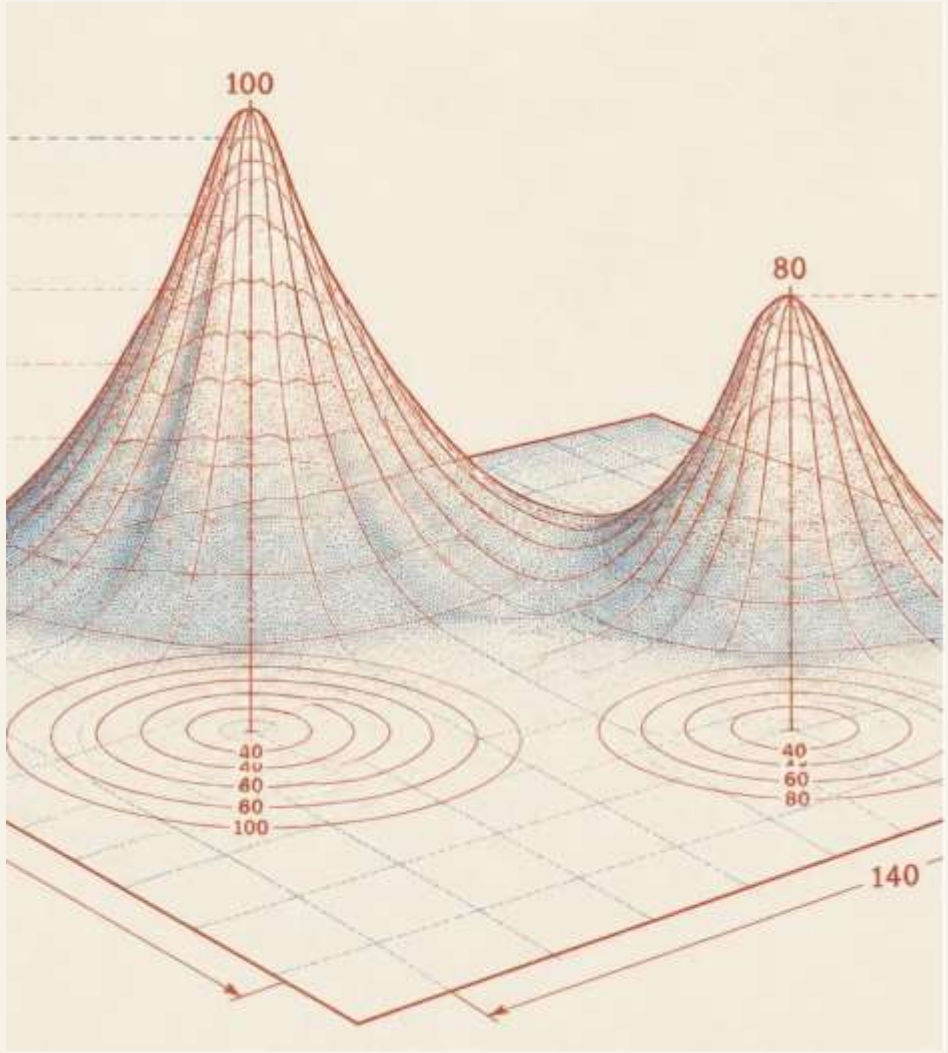
Evaluate accumulated exposure and average strength along walking path.

Computational toolkit: meshgrid, surf, contour, gradient, quiver, trapz in GNU Octave / MATLAB · 04 / 10

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The signal landscape: two peaks and a deep valley

- Surface plot shows Router 1 (100) and Router 2 (80) as peaks
- Contours ripple outward from each router
- Signal falls to near zero at the far corners



3D surface plot with contour projection (Figure 1).

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Gradients point to the routers and the path crosses a trough

- Gradient arrows point toward the peaks and shrink to near zero at them.
- Student path x = 1 + 8t, y = 9 − 8t; length 8√2 ≈ 11.3 m.
- The path slices through the weak zone between both routers.
- Gradients show which way to move a router.

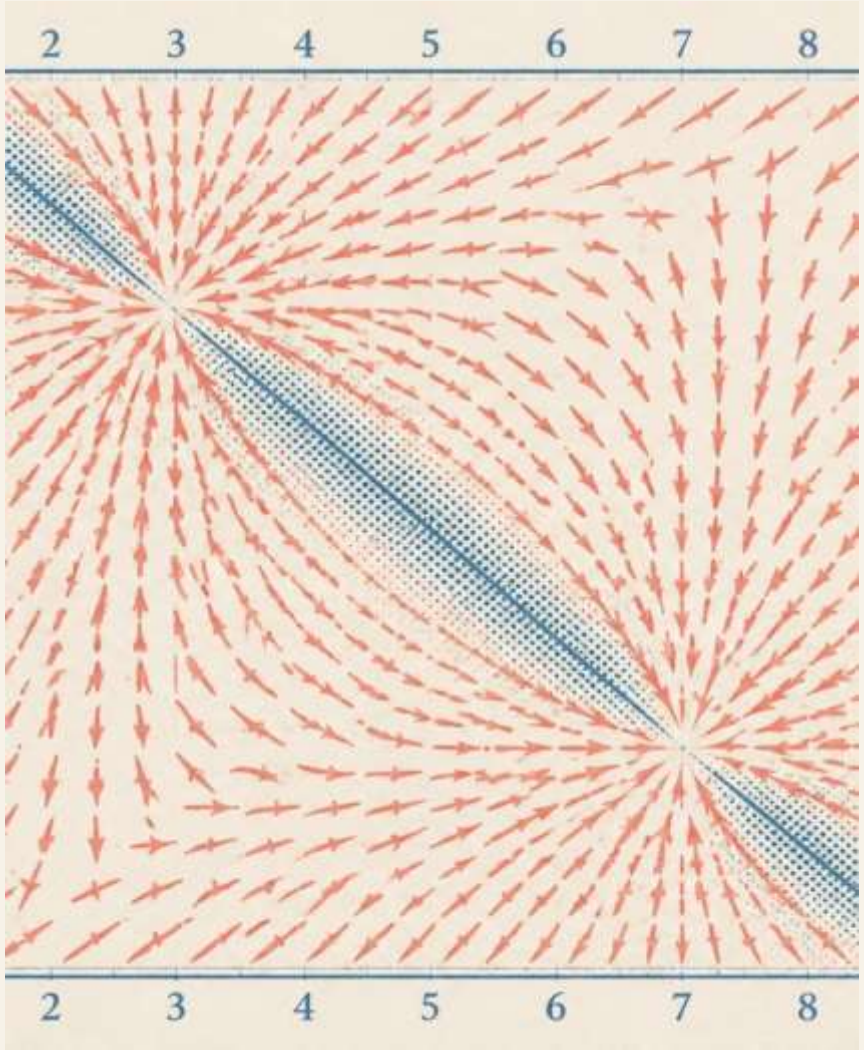


Figure 2 · Contour map with gradient arrows and the path

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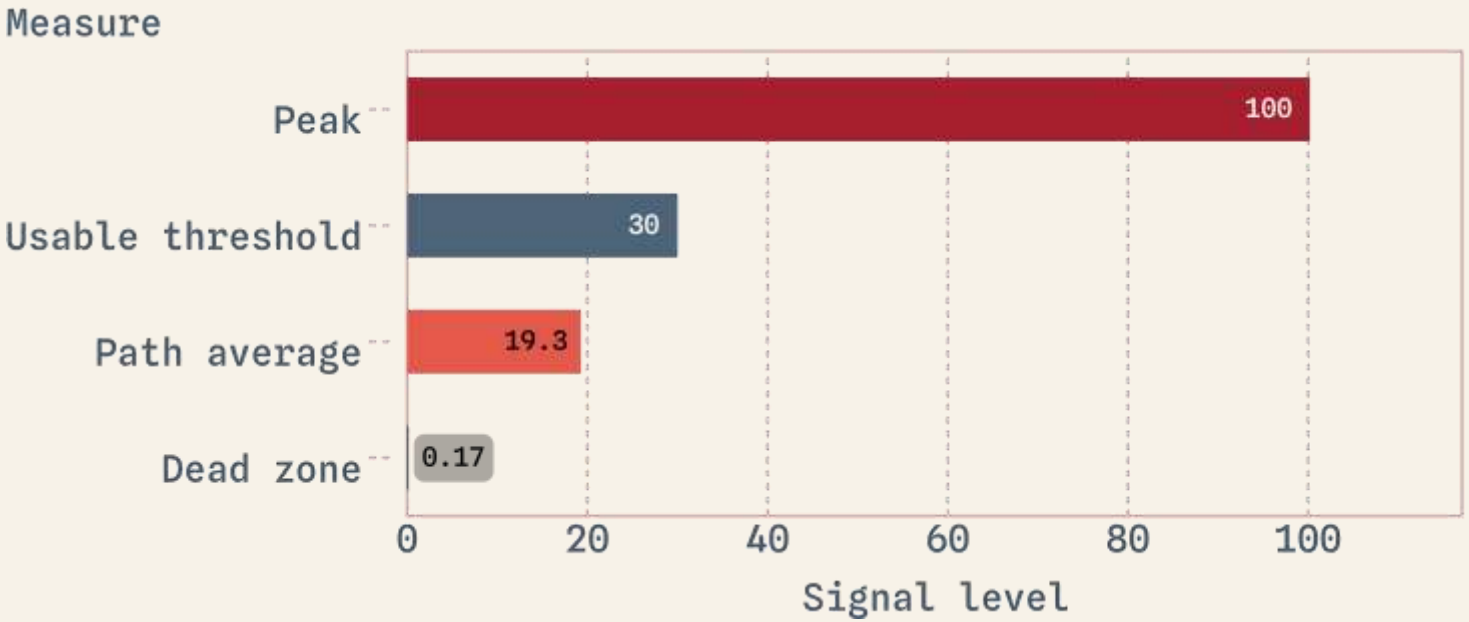
The numbers: 100 at the peak, 0.17 in the corner

100.18
Peak signal at (2,2)

0.17
Dead zone at (0,10)

60.76
Usable area (S > 30)

19.27
Path mean strength



Integration metrics

Usable coverage (S > 30): 60.76 of 100 m^2, about 61%.

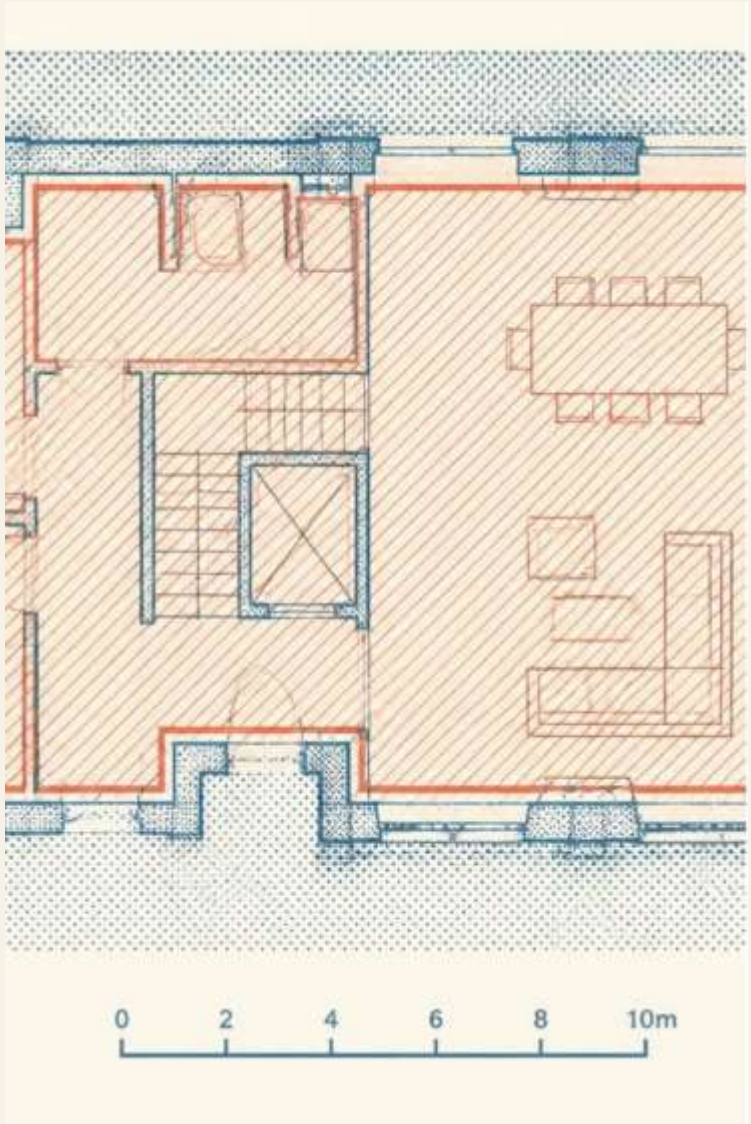
Line integral: $\int S\, ds = 218.05$ along path.

Volume integral: $\iint S\, dA = 3{,}995.59$ total units.

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What the math gives us: clear, measurable answers

- Makes the invisible signal visible
- Pinpoints strongest and weakest spots
- Quantifies coverage: 61% usable
- Simple, reusable class-level Octave code



S > 30 working zone

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Where the model falls short: 39% dark and a weak path

39%

- About 39 m^2 of the room is below the usable level
- Walking path averages 19, under the threshold of 30
- Model is assumed, ignoring walls, furniture and interference
- Threshold of 30 is a chosen value; integrals are grid approximations

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The path forward: move, measure, mesh

01 Relocate Router 2

Shift coordinate from (8,7) toward (4,8) to reinforce the starved (0,10) quadrant.

02 Re-run script

Evaluate delta in usable area and verify transit path average clears threshold 30.

03 Empirical calibration

Collect RSSI dBm probe samples across room to calibrate Gaussian decay variance.

04 Model obstructions

Incorporate attenuation line integrals for structural partitions, glass and furniture.

05 Gradient ascent

Automate optimal node placement via multivariate gradient ascent on path integral.

Conclusion: vector calculus transforms intuition into deterministic radio engineering · 10 / 10

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Signal model

$$S(x,y) = 100\,e^{-\frac{(x-2)^2+(y-2)^2}{10}} + 80\,e^{-\frac{(x-8)^2+(y-7)^2}{10}}$$

Router 1 at (2, 2)

First term of S(x, y):
100·e^(-((x-2)² + (y-2)²)/10)

Router 2 at (8, 7)

Second term of S(x, y):
80·e^(-((x-8)² + (y-7)²)/10)

10 m × 10 m room

Each router gives a strong signal nearby that fades with distance.

In Octave:

```
[x,y] = meshgrid(0:0.1:10, 0:0.1:10);
S = 100*exp(-((x-2).^2 + (y-2).^2)/10) + 80*exp(-((x-8).^2 + (y-7).^2)/10);
```

Signal model: S(x, y) = 100·e^(-((x-2)² + (y-2)²)/10) + 80·e^(-((x-8)² + (y-7)²)/10), with Router 1 at (2, 2) and Router 2 at (8, 7).

Wi-Fi Signal Mapping and Router Placement

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What we do

01

Exp 2

3D surface and contour plot of the signal

02

Exp 3

Strongest spot and dead zone (maxima / minima)

03

Exp 5

Usable coverage area and total signal (double integral)

04

Exp 7

Gradient arrows: direction of fastest signal increase

05

Exp 8

Total signal along a walking path (line integral)

Using the calculus topics from our lab experiments, we then analyse the room in Octave.

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2. Results

61 m²

usable signal (above 30) out of 100 m²

- Strongest signal is about 100, at Router 1 (2, 2).
- The dead zone is at the corner (0, 10), where the signal is almost 0.
- About 61 m² of the 100 m² room has usable signal (above 30).
- The walking path from (1, 9) to (9, 1) averages a weak signal of about 19, so the routers should be repositioned.

Note: The signal formula is an assumed model, not real measurements. The code is simple and uses only what we learned in class (meshgrid, surf, contour, gradient, quiver, trapz).

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3. Octave Code (Input) · 1 of 3

```
clc; clear; close all;          % clear screen, variables and figures
pkg load symbolic               % load the symbolic package (needed for syms and int)

% ---------- Room and routers ----------
% Room is 10 m x 10 m. Router 1 is at (2,2), Router 2 is at (8,7)
[x,y] = meshgrid(0:0.1:10, 0:0.1:10);   % grid of points covering the room (step 0.1 m)

% Signal = sum of two bell-shaped signals, one from each router
% Router 1 has peak strength 100, Router 2 has peak strength 80
S = 100*exp(-((x-2).^2 + (y-2).^2)/10) + 80*exp(-((x-8).^2 + (y-7).^2)/10);

% ---------- 1. 3D surface of the signal (Exp 2) ----------
figure(1);                      % open figure window 1
surf(x, y, S);                  % draw the signal as a 3D surface
shading interp;                 % smooth colours on the surface
colorbar;                       % show the colour scale
xlabel('x (m)'); ylabel('y (m)'); zlabel('Signal strength');   % axis labels
title('Wi-Fi signal surface');  % plot title
% ---------- 2. Strongest and weakest point (Exp 3) ----------
[Smax, i] = max(S(:));          % highest signal value and its position number
[Smin, j] = min(S(:));          % lowest signal value and its position number
disp(['Strongest signal = ', num2str(Smax), ' at (', num2str(x(i)), ', ', num2str(y(i)), ')']);   % print strongest point
disp(['Weakest signal   = ', num2str(Smin), ' at (', num2str(x(j)), ', ', num2str(y(j)), ')']);   % print weakest point
```

Wi-Fi Signal Mapping and Router Placement

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3. Octave Code (Input) - 2 of 3

```
% ---------- 3. Coverage ----------
% Usable area: we only count the grid boxes where signal > 30
dx = 0.1; dy = 0.1;            % size of each small grid box
usable = S > 30;               % 1 where signal is above 30, otherwise 0
area = sum(usable(:)) * dx * dy;       % number of usable boxes x box area = usable area (m^2)
disp(['Usable area (signal > 30) = ', num2str(area), ' m^2 out of 100 m^2']);   % print usable area

% Total signal: double integral using int twice (Exp 5)
syms X Y                       % symbolic variables (capital, so they do not clash with the grid x, y)
S1 = 100*exp(-(X-2)^2/10)*exp(-(Y-2)^2/10);    % signal of Router 1 only
S2 = 80*exp(-(X-8)^2/10)*exp(-(Y-7)^2/10);     % signal of Router 2 only
% inner int: x from 0 to 10, outer int: y from 0 to 10, for each router, then add
total = int(int(S1, X, 0, 10), Y, 0, 10) + int(int(S2, X, 0, 10), Y, 0, 10);
disp(['Total signal over the room = ', num2str(double(total))]);   % double() turns the symbolic answer into a number

% ---------- 4. Gradient arrows (Exp 7) ----------
[gx, gy] = gradient(S, 0.1);    % gx = dS/dx and gy = dS/dy at every point

figure(2);                     % open figure window 2
contour(x, y, S, 15);          % contour lines (15 levels) of equal signal
hold on;                       % keep the contour plot and add more on top
colorbar;                      % show the colour scale
```


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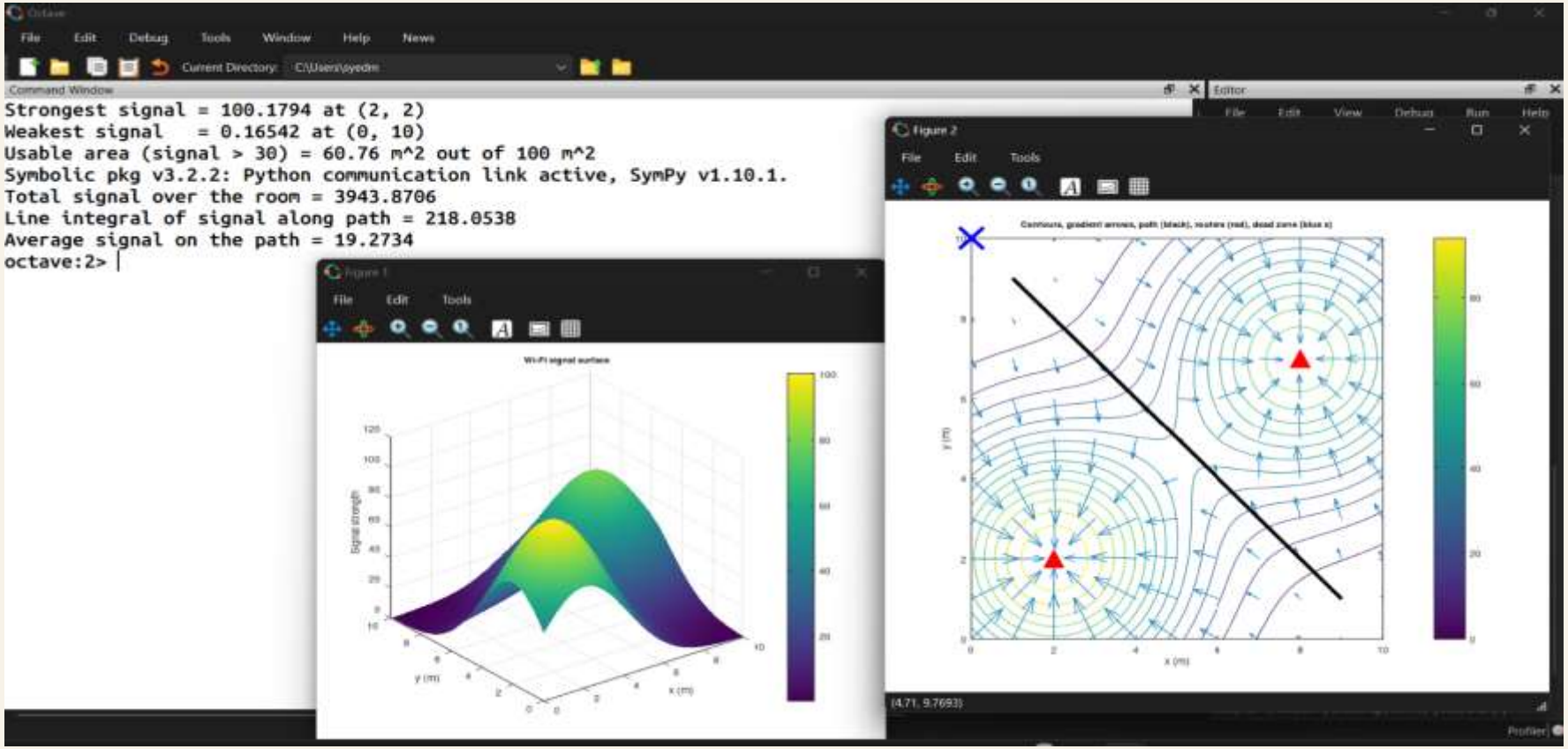
3. Octave Code (Input) - 3 of 3

```
% draw gradient arrows (every 10th point so the plot is not crowded)
quiver(x(1:10:end,1:10:end), y(1:10:end,1:10:end), gx(1:10:end,1:10:end), gy(1:10:end,1:10:end));
% ---------- 5. Walking path: line integral (Exp 8) ----------
% Path from hostel (1,9) to class (9,1) is a straight line: x = 1+8t, y = 9-8t
t = 0:0.01:1;                % t goes from 0 to 1 (used only to draw the path)
px = 1 + 8*t;                % x-coordinates of the path
py = 9 - 8*t;                % y-coordinates of the path
% Line integral using the midpoint sum
dt = 0.01;                   % width of each small piece of t
tm = 0.005:0.01:0.995;       % midpoint of each of the 100 pieces
pxm = 1 + 8*tm;              % x-coordinates at the midpoints
pym = 9 - 8*tm;              % y-coordinates at the midpoints
% signal at every midpoint of the path
Sp = 100*exp(-((pxm-2).^2 + (pym-2).^2)/10) + 80*exp(-((pxm-8).^2 + (pym-7).^2)/10);
len = 8*sqrt(2);             % length of the path = sqrt(8^2 + 8^2)
line_int = sum(Sp) * dt * len;  % add (signal x small piece of length) over the path
disp(['Line integral of signal along path = ', num2str(line_int)]);   % print line integral
disp(['Average signal on the path = ', num2str(line_int/len)]);      % total / length = average signal
% ---------- Marks on the plot ----------
plot(px, py, 'k', 'LineWidth', 2);     % black line = walking path
plot(2, 2, 'r^', 'MarkerSize', 10, 'MarkerFaceColor', 'r');   % red triangle = Router 1
plot(8, 7, 'r^', 'MarkerSize', 10, 'MarkerFaceColor', 'r');   % red triangle = Router 2
plot(x(j), y(j), 'bx', 'MarkerSize', 12, 'LineWidth', 2);     % blue X = dead zone (weakest point)
xlabel('x (m)'); ylabel('y (m)');      % axis labels
title('Contours, gradient arrows, path (black), routers (red), dead zone (blue x)');   % plot title
```

Wi-Fi Signal Mapping and Router Placement

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4. Output · Screenshot



Screenshot (editor, Command Window, Figure 1 and Figure 2):

Wi-Fi Signal Mapping and Router Placement

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5. Conclusion

- Figure 1 shows two peaks (the routers) and weak regions near the edges.
- Figure 2 shows that the gradient arrows point towards the routers, which is the direction in which the signal increases fastest.
- The arrows are short near the routers because the gradient is almost zero at a peak.
- The black walking path crosses the weak zone between the two routers, so adding or moving a router would improve coverage along that path.

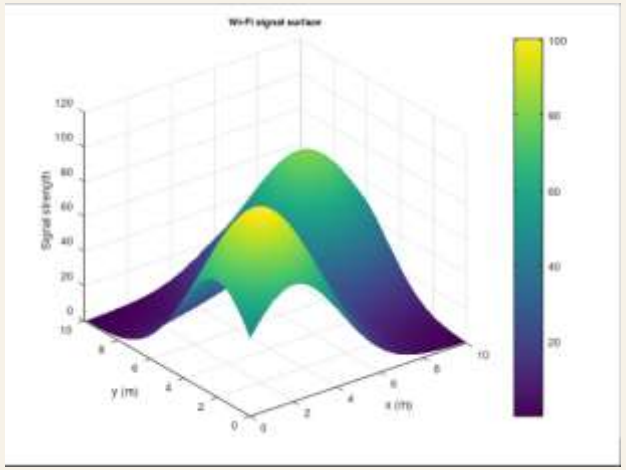


Figure 1 · Wi-Fi signal surface

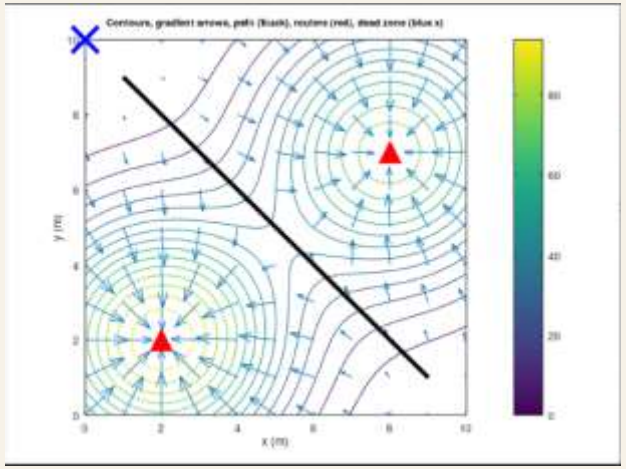


Figure 2 · Contours, gradient arrows, path, routers

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